$i_C = C\cdot\frac{dU_C}{dt}$, $i_2 = \frac{U_C}{4\Omega} = \frac{1}{4}U_C$

$$\boxed{i_C + i_2 = C\cdot\frac{dU_C}{dt} + \frac{1}{4}U_C \quad -①}$$

$U_L = L\cdot\frac{di_L}{dt}$

$i_1 = \frac{U_L}{6\Omega} = \frac{1}{6}L\cdot\frac{di_L}{dt}$

$\cancel{i_L + i_L}$

$$\boxed{i_L + i_1 = i_L + \frac{1}{6}L\cdot\frac{di_L}{dt} \quad -②}$$

$\frac{1}{4}U_C + C\cdot\frac{dU_C}{dt} = 1$

$\frac{1}{4}U_C + \frac{1}{8}\cdot\frac{dU_C}{dt} = 1$

$\frac{1}{8}\cdot\frac{dU_C}{dt} = 1 - \frac{1}{4}U_C$

$\frac{dU_C}{dt} = 8 - 2U_C$

$= 2(4 - U_C)$

$\frac{dU_C}{4-U_C} = 2dt$

$\frac{-d(4-U_C)}{4-U_C} = 2dt$

$-\ln|4-U_C|\Big|_{10}^{t} = 2t$

$\ln(U_C-4)\Big|_{10}^{U} = -2t$

$\ln(U_t-4) - \ln 6 = -2t$

$\ln(U_t-4) = -2t + \ln 6$

$\cancel{\ln(U_t} - 4$

$U_t - 4 = 6e^{-2t}$

$U_t = 6e^{-2t} + 4$

$i_L + \frac{1}{6}L\cdot\frac{di_L}{dt}$

$i_L + \frac{1}{3}\cdot\frac{di_L}{dt} = 1$

$\frac{1}{3}\cdot\frac{di_L}{dt} = 1 - i_L$

$\frac{di_L}{1-i_L} = 3dt$

$\frac{d(1-i_L)}{1-i_L} = -3dt$

$\ln|1-i_L|\Big|_0^{i_L} = -3t$

$\ln(1-i_L)\Big|_0^{i_L} = -3t$

$\ln(1-i_L) = -3t$

$1 - i_L = e^{-3t}$

$i_L = 1 - e^{-3t}$

$\frac{di_L}{dt} = 3e^{-3t}$

$L\cdot\frac{di_L}{dt} = 6e^{-3t}$